Executive Project Report
Automobile Parts Sales & Operational Analytics

Primary Tool: PostgreSQL

Domain: E-Commerce & B2B Automobile Parts Supply Chain

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Tools Used: PostgreSQL, DBeaver, CTEs, Windows Functions, SubQueries

GitHub: Click Here

1. Executive Summary & Project Overview

This project delivers a comprehensive data-driven evaluation of an international automobile parts supply chain. Utilizing an advanced PostgreSQL relational database, the analysis processes multi-table transactional data to uncover critical insights across sales performance, customer behavior, logistics efficiency, and supply chain bottlenecks.

Key strategic findings reveal that the **Dealer channel** is nearly 15 times more effective than online and retail channels, **South Asia** dominates regional revenue generation, and structured operational inefficiencies - such as an 8% discount leakage and a 4% cancellation rate - present immediate avenues for profitability optimization.

2. Methodology & Analytical Frameworks

The analysis was executed using complex PostgreSQL scripts leveraging modern SQL capabilities, including **Common Table Expressions (CTEs), Window Functions (NTILE, RANK, ROW_NUMBER), Multi-Table Joins, and Aggregate Functions**.

The core evaluation modules covered:

- **Financial & Sales Analysis:** Month-over-Month (MoM) & Year-over-Year (YoY) growth, Average Order Value (AOV), top/bottom revenue products, OEM vs. Non-OEM production distribution, and promotion effectiveness.
- **Customer Intelligence:** Recency, Frequency, Monetary (RFM) segmentation, Customer Lifetime Value (CLV), Customer Cohort Retention, and Dealer Credit Limit analysis.
- **Operational & Supply Chain Efficiency:** Order funnel conversion tracking, cycle time, average shipping (2 days) vs. delivery (8 days) timelines, warehouse utilization, and return/refund root-cause analysis.

3. Core Findings & Business Insights

A. Channel Performance & Regional Dynamics

- **Channel Dominance:** The **Dealer channel** significantly outperforms online and retail channels, generating nearly **15 times** the revenue volume. However, existing acquisition channels are largely underperforming in direct product conversion.
- **Regional Leaders: South Asia** emerges as the primary revenue-generating region, spearheaded by top-performing sales representative **Michael Richardson** and bolstered by robust regional demand. Conversely, **Sub-Saharan Africa (Retail)** experiences the highest

concentration of delayed orders, while **Eastern Europe (Online)** and **Latin America (Distribution Center 9)** record the highest order return rates.

- **Warehouse Optimization: North Africa Distribution Center 5** stands out as the most efficiently utilized logistics hub in the network.

B. Financial Leakages & Pricing Impact

- **Revenue Leakage:** The business faces a **4% total revenue loss due to order cancellations** and an **8% total revenue loss due to discounting strategies**.
- **Category & Product Drivers:**
    - **Premium car batteries** generate the highest overall revenue.
    - **Spark plugs** within the electrical category deliver exceptionally high profit margins.
    - **Engine mouths (Online)** suffer as the most frequently returned product category, driven predominantly by **duplicate orders**.
- **Production Distribution:** The organization produces products for **non-OEM applications nearly 3 times more** than OEM parts, though products tied to **Tata Terra** secure the highest individual revenue yields. Furthermore, the top **80% of total revenue** is concentrated across **9,531 individual auto parts**.

C. Customer Segmentation & Retention

- **Tier Performance:** Bronze-tier dealers are proving exceptionally strong in driving consistent revenue flow.
- **CLV & Intermediate Segments:** A significant concentration of customers resides within the intermediate CLV segment. Transitioning these intermediate customers into loyal tiers represents a major untapped growth opportunity.

4. Strategic Business Recommendations

Based on the PostgreSQL findings, leadership should implement the following strategic roadmap:

1. **Reallocate Marketing & Sales Capital to Dealer Networks:** Given that the dealer channel outperforms online and retail channels by 15x, expand credit limits for high-performing Bronze-tier dealers and scale regional support in South Asia.

2. **Mitigate Revenue Leakages (Discounts & Cancellations):** Restructure discount approval workflows to curb the 8% revenue loss, and introduce automated duplicate-order detection at checkout to address the primary driver of product returns (especially for online engine mouths).
3. **Optimize Logistics in Bottleneck Regions:** Audit shipping protocols in Sub-Saharan Africa to resolve order delays, and investigate quality control or description inaccuracies in Eastern Europe and Latin American distribution hubs to lower return rates.
4. **Targeted Customer Lifecycle Campaigns:** Develop automated email and CRM workflows specifically designed to nudge intermediate-tier customers into higher loyalty bands, maximizing Customer Lifetime Value (CLV).
5. **Strategic Product Alignment:** Re-evaluate the 3:1 non-OEM to OEM production ratio, focusing manufacturing capacity on high-margin electrical parts (like spark plugs) and top revenue-producing lines (like premium car batteries and Tata Terra compatible parts).